SUPERSTORE SALES ANALYTICS

Project Planning & Requirements Document

Dataset: Superstore Sales Dataset

Project Team

Mahmoud — Data Cleaning

Abdelrahman Ramadan — DAX & Measures

Amr & Ramz — Data Visualization

Abdelrahman Farouk — Reporting

Prepared for project planning and implementation

1. Project Overview

The Superstore Sales Analytics project aims to transform the provided sales dataset into a clear, interactive Power BI report. The report will help users understand sales performance across time, products, customer segments, and geographic regions, and explore shipping patterns using the available order and shipping dates.

2. Project Objectives

- Build a reliable and well-structured dataset through data profiling, cleaning, and validation.
- Create consistent DAX measures for key sales KPIs and time-based analysis.
- Design an intuitive interactive dashboard with clear visuals, filters, and drill-down capabilities.
- Provide a reporting view that summarizes the most important findings and supports business decisions.
- Answer at least 40 business questions using the fields available in the dataset.
- Document assumptions, data limitations, and the calculation logic used in the report.

3. Dataset Summary

The supplied CSV contains 9,800 rows and 18 columns. The fields include order identifiers and dates, shipping mode, customer details, segment, country, city, state, region, product identifiers, category, sub-category, product name, and sales.

Data Area	Available Fields	Potential Analysis
Orders & Time	Order ID, Order Date, Ship Date, Ship Mode	Sales trends, order counts, shipping duration
Customers	Customer ID, Customer Name, Segment	Sales by segment and customer
Geography	Country, City, State, Postal Code, Region	Sales by region, state, and city
Products	Product ID, Category, Sub-Category, Product Name	Sales by category, sub-category, and product
Sales	Sales	Total sales, average sales per order, sales contribution

4. Scope & Key Deliverables

- Clean and validated source data in Power Query.
- A documented data model and date table, where appropriate.
- A set of reusable DAX measures for sales and order analysis.
- An interactive Power BI dashboard with KPI cards, trend charts, product analysis, and geographic analysis.
- A concise report page or summary of key findings and recommendations.
- A requirements checklist showing which business questions are answered by the final report.

5. Project Stages & Tasks

Stage	Main Tasks	Owner(s)	Output
1. Planning & Requirements	Confirm objectives, define KPIs, list business questions, agree on report layout and scope.	Whole team	Approved requirements document
2. Data Profiling & Cleaning	Inspect column types, check missing values and duplicates, standardize dates/text, validate sales values, document cleaning decisions.	Mahmoud	Clean dataset and cleaning log
3. Data Modeling & DAX	Create relationships if needed, build a date table, define measures, validate calculations against source data.	Abdelrahman Ramadan	Data model and DAX measure set
4. Visualization	Build KPI cards, trend charts, category/product	Amr & Ramz	Interactive dashboard

	visuals, regional visuals, slicers, drill-through and interactions.		pages
5. Reporting & Insights	Summarize findings, highlight trends and top/bottom performers, write evidence-based insights and recommendations.	Abdelrahman Farouk	Executive summary/report
6. Testing & Final Delivery	Check filters, totals, date logic, visual interactions, readability, and consistency; fix issues and prepare presentation.	Whole team	Tested final Power BI file and presentation

6. Team Roles & Responsibilities

Team Member	Assigned Role	Responsibilities
Mahmoud	Data Cleaning	Profile the CSV; handle missing or invalid values; check duplicates; set correct data types; standardize date and text fields; document every transformation.
Abdelrahman Ramadan	DAX & Measures	Create and validate measures such as Total Sales, Order Count, Average Sales per Order, Sales by Period, and Shipping Days; document measure definitions.
Amr	Data Visualization	Design dashboard layout; choose suitable visuals; build KPI cards, trend charts, and product/geographic analysis; ensure visual clarity and consistent formatting.
Ramz	Data Visualization	Work with Amr on visual development; configure slicers and cross-filtering; test interactions; improve navigation and user experience.
Abdelrahman Farouk	Reporting	Prepare the executive summary; describe major trends and top/bottom performers; connect findings to business questions; provide cautious, evidence-based recommendations.

7. Tools & Technologies

Tool	Purpose
CSV / Microsoft Excel	Source review, quick checks, and validation of sample totals.
Power Query	Data profiling, cleaning, transformation, and repeatable preparation steps.
Power BI Desktop	Data modeling, DAX, interactive report design, and final dashboard delivery.
DAX	Measures, KPI calculations, time analysis, and dynamic aggregations.
Microsoft Word	Requirements documentation and final report writing.
PowerPoint (optional)	Final presentation of project approach, dashboard, and findings.

8. Business Questions / Requirements

The following questions define the minimum analytical requirements. The team should map each question to one or more visuals, measures, or report elements. Questions requiring fields not present in the supplied CSV are marked as future-data requirements.

ID	Business Question / Requirement	Suggested Output	Data Status
01	What is the total sales revenue for the selected period?	KPI card	Available
02	How do total sales change month over month?	Line chart	Available
03	How do total sales change year over year?	Line chart / KPI	Available, subject to date coverage
04	Which year has the highest total sales?	Column chart / KPI	Available
05	Which month has the highest total sales?	Monthly trend chart	Available
06	Which month has the lowest total sales?	Monthly trend chart	Available
07	How many distinct orders are recorded?	KPI card	Available
08	What is the average sales value per order?	KPI card	Available; define order-

			level calculation
09	How are sales distributed across customer segments?	Bar / donut chart	Available
10	Which customer segment contributes the most sales?	Ranked bar chart	Available
11	Which customer segment contributes the least sales?	Ranked bar chart	Available
12	What is the sales contribution percentage of each segment?	100% stacked / table	Available
13	Which region generates the highest sales?	Bar chart / map	Available
14	Which region generates the lowest sales?	Bar chart / map	Available
15	How do sales compare across all regions?	Clustered bar chart	Available
16	Which states have the highest sales?	Ranked bar chart / map	Available
17	Which states have the lowest sales?	Ranked bar chart	Available
18	Which cities generate the highest sales?	Ranked bar chart	Available
19	How are sales distributed geographically?	Map / filled map, if supported	Available
20	Which product category generates the highest sales?	Bar chart	Available
21	Which product category generates the lowest sales?	Bar chart	Available
22	What percentage of total sales comes from each category?	Donut / 100% stacked chart	Available
23	Which sub-categories generate the highest sales?	Ranked bar chart	Available
24	Which sub-categories generate the lowest sales?	Ranked bar chart	Available
25	Which products have the highest sales value?	Top-N bar chart / table	Available
26	Which products have the lowest sales value?	Bottom-N table	Available; interpret carefully
27	What are the top 10 products by sales?	Top-N bar chart	Available
28	How do category sales change over time?	Line chart with category legend	Available
29	How do sub-category sales compare within each category?	Matrix / bar chart	Available
30	Which products contribute most to sales within each category?	Drill-down bar chart	Available
31	How many distinct customers are represented in the dataset?	KPI card	Available
32	Which customers have the highest total sales?	Top-N customer table	Available
33	What share of sales is generated by the top 10 customers?	KPI / Pareto chart	Available
34	How concentrated are sales among the highest-value customers?	Cumulative contribution / Pareto chart	Available
35	Which shipping mode is used most frequently by order line?	Bar chart	Available; clarify row vs order count
36	How much sales value is associated with each shipping mode?	Bar chart	Available
37	What is the average number of days between order date and ship date?	KPI card	Available after date validation
38	Which shipping modes have the longest average shipping duration?	Bar chart	Available
39	How does shipping duration vary by region?	Bar chart / matrix	Available
40	Are there months with high sales and longer average shipping duration?	Combo chart	Available
41	How do sales differ by weekday or month?	Trend / heatmap-style matrix	Available after date fields are parsed
42	Which region leads in sales for each product category?	Matrix / small multiples	Available
43	Which customer segment leads in each product category?	Matrix / stacked chart	Available
44	What are the top product sub-categories in each region?	Matrix / Top-N chart	Available
45	How do selected products perform across regions?	Interactive matrix	Available
46	Can users filter the report by date, region, segment, category, and shipping mode?	Slicers	Required report functionality
47	Can users drill from category to sub-category and then product?	Drill-down hierarchy	Required report functionality
48	Can users view detailed sales records behind a summary visual?	Detail table / drill-through	Required report functionality
49	Are sales totals consistent between the cleaned data and Power BI measures?	Validation check	Required quality control
50	Are date filters and visual interactions working consistently across report pages?	Report testing checklist	Required quality control
51	What is the total profit and profit margin by category?	Profit KPI / category chart	Future requirement: Profit

			field is not in this CSV
52	Which products have the highest discount and how does discount affect sales?	Discount analysis	Future requirement: Discount field is not in this CSV
53	How many units of each product were sold?	Quantity chart	Future requirement: Quantity field is not in this CSV
54	What is the cost and gross margin for each product?	Cost / margin analysis	Future requirement: Cost and profit fields are not in this CSV
55	Which orders were returned or cancelled?	Returns analysis	Future requirement: return/cancellation data is not in this CSV

9. Proposed Dashboard Pages

Page	Purpose	Suggested Components
1. Executive Overview	Provide a quick summary of overall performance.	Total Sales, Order Count, Customer Count, Average Sales per Order, monthly sales trend, sales by region, sales by category.
2. Product Analysis	Compare product and category performance.	Sales by category/sub-category, Top 10 products, product detail table, category drill-down.
3. Customer & Segment Analysis	Understand who contributes to sales.	Sales by segment, top customers, customer contribution, segment/category matrix.
4. Geographic Analysis	Identify strong and weak markets.	Sales by region/state/city, ranked locations, region/category matrix.
5. Shipping Analysis	Explore shipping mode and delivery timing patterns.	Orders by ship mode, sales by ship mode, average shipping days, shipping duration by region.
6. Details & Validation (optional)	Support detailed investigation and quality checks.	Order-level detail table, filter controls, validation notes, data definitions.

10. Proposed KPI Definitions

KPI	Suggested Definition	Important Note
Total Sales	Sum of the Sales column.	Use the cleaned numeric Sales field.
Order Count	Distinct count of Order ID.	Do not count rows as orders because one order may contain multiple product lines.
Distinct Customers	Distinct count of Customer ID.	Customer names may not be a unique identifier.
Average Sales per Order	Total sales divided by distinct Order ID count.	This is an order-level average, not average row sales.
Average Shipping Days	Average of Ship Date minus Order Date.	Validate date parsing and investigate negative or missing durations.
Sales Contribution %	Sales for the selected group divided by total sales in the comparison context.	Document how slicers affect the denominator.

11. Data Quality Checks & Assumptions

- Confirm that Order Date and Ship Date are parsed as dates in the correct day/month format; validate several sample records before applying transformations.
- Check missing values in Postal Code and other columns. Do not remove records automatically without assessing impact.

- Check duplicates using the correct grain. Repeated Order ID values can be valid because an order can contain multiple product lines.
- Ensure Sales is numeric and investigate null, negative, or unusually large values.
- Use Order ID for order counts and Customer ID for distinct customer counts.
- Document all Power Query transformations so the cleaning process can be reproduced.
- The supplied dataset contains Sales but does not contain Profit, Quantity, Discount, Cost, Returns, or a target delivery date. Do not claim to calculate these metrics unless additional data is provided.
- Treat patterns as descriptive insights. Sales alone cannot prove profitability, discount effectiveness, or the cause of a performance change.

12. Acceptance Criteria

- The cleaned dataset refreshes without errors and has documented transformation steps.
- Core KPI values are checked against the source data or an independent calculation.
- At least 40 business questions are documented; the final report addresses the applicable questions supported by the data.
- The report includes clear titles, consistent number formats, readable labels, and sensible visual choices.
- Date, region, segment, category, sub-category, and shipping-mode filters behave as expected where applicable.
- The report avoids unsupported claims and clearly identifies questions that need additional data.
- Each team member's assigned work is reviewed and integrated into one final Power BI report.

13. Suggested Final Submission Package

- Cleaned dataset or Power Query steps.
- Power BI (.pbix) file.
- This Project Planning & Requirements Document.
- A short executive summary containing key findings and recommendations.
- A final presentation demonstrating the dashboard and explaining each team member's contribution.